INCH-POUND

MIL-DTL-25666/6 (USAF)
15 September 1995
SUPERSEDING
MIL-C-27652D (USAF)
04 December 1980

DETAIL SPECIFICATION

CONVERTER, LIQUID OXYGEN GCU-20/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of converter. It provides storage and pressurization of 75 liters of liquid oxygen to 300 pounds per square inch gauge (psig), nominal pressure and does not include warming coils. The converter is designated GCU-20/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index for Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-DTL-25666	Converters, Liquid Oxygen, Capacitance Type Gauging, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-3037 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of the above specifications, and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The connectors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design and construction. The converter shall conform to figures 1 through 3. The converter shall have an operating pressure of 300 pounds per square inch gage (psig) and a minimum capacity of 75 liters of liquid oxygen measured at 1 atmosphere plus at least 3 percent of this volume to allow for the change in liquid density at the operating pressure,

3.2.1 Strength. The filled converter shall be designed to withstand the dynamic loads that are imposed by the following accelerations:

Vertical down 9G

Vertical up 3G

Lateral 4G

3.2.2 Marking of ports. The ports shall be marked as indicated in figure 2.

3.2.3 Pressure relief valves. The pressure relief valves shall conform to types III and IV of MIL-V-9050 and shall be of such size and shape to fit the over-all dimensions specified for the converter.

3.2.4 Weight. The weight of the converter shall not exceed 80 pounds.

3.3 Performance.

3.3.1 Final assembly pressure. The final assembly pressure of the converter shall be 320 psig.

3.3.2 Filling time. The time required to fill the converter shall not exceed 30 minutes.

3.3.3 Evaporation loss.

3.3.3.1 Vented condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed 9 pounds.

3.3.3.2 Buildup condition. The evaporation loss of the essentially full converter in the buildup condition shall not exceed 5 pounds.

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3.3.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed 15 minutes.

3.3.5 Capacitance. The capacitance of the sensing element, after it is installed in the empty converter, shall be 910.0 ±7.5 picofarad (pf).

3.3.6 Inner vessel pressure.

3.3.6.1 Inner vessel. The inner vessel shall meet the performance requirements for the inner vessel pressure test specified in MIL-DTL-25666 when subjected to a pressure of 500 psig.

3.3.6.2 Permanent volumetric expansion. The converter shall meet the performance requirements for permanent volumetric expansion as specified in MIL-DTL-25666 when subjected to a pneumatic or hydrostatic pressure of 500 psig .

3 3.7 Delivery rate.

3.3.7.1 Constant flow. The converter shall deliver a flow of gaseous oxygen up to 400 liters per minute for not less than 2.25 hours with the operating pressure remaining between 295 and 315 psig after the first 15 minutes of flow. An intermittent increase or decrease of pressure by an additional 5 psig is allowable, but these variations shall not exceed a 2-minute interval for any one pressure cycle. The flow at the specified rate shall be initiated within 10 minutes after the converter has reached the operating pressure. The full test procedure is specified in MIL-DTL-25666.

3.3.8 Calibration. The capacitance sensing element shall have characteristics that provide for a linear relationship between electrical capacitance and volume of liquid oxygen sensed. The relationship shall be in accordance with figure 4, using the equations that are specified (see 3.3.8.1) to determine tolerances. An example of the use of these equations is as follows: If the actual quantity of liquid oxygen at the 30-liter test point is 30.2 liters, the measured capacitance must be between C maximum and C minimum.

C minimum = 5.725 x 30.2 + 900.6 = 1073.5 pf
C maximum = 5.875 x 30.2 + 919.4 = 1096.8 pf

3.3.8.1 Equations. The following equations shall be used to calculate the maximum and the minimum capacitances.

Values less than 15 liters	Values greater than 15 liters
C minimum = 5.6L + 902.5	C minimum = 5.725L + 900.6
C maximum = 6L + 917.5	C maximum = 5.875L + 919.4

C = Capacitance in pf
L = Exact quantity in liters of liquid at test point

3.3.9 Operating temperature. The operating pressure shall remain between 295 and 320 psig after the first 15 minutes of flow at the low temperature and shall remain between 290 and 315 psig after the first 15 minutes of flow at the high temperature. An intermittent increase or decrease of pressure by an additional 5 psig is allowable provided these variances do not exceed a 2-minute interval for any one pressure cycle.

3.3.10 Vibration.

3.3.10.1 Vibration. There shall be no evidence of damage to the converter as a result of these tests.

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3.3.10.2 Vibration - resonance. The converter shall not exhibit physical evidence of damage after completion of the resonance portion of the vibration test specified in MIL-DTL-25666.

3.3.10.3 Vibration - cycling. The converter shall not exhibit physical evidence of damage after completion of the cycling portion of the vibration test specified (see 4.4.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The qualification inspection shall consist of all the inspection methods specified in MIL-DTL-25666.

4.3 Conformance inspection. The conformance inspection shall consist of the individual tests and the sampling plans and tests specified in MIL-DTL-25666.

4.4 Vibration.

4.4.1 Vibration - cycling. The converter shall be vibrated under the cycling conditions specified herein for the applicable periods listed in table I. The frequency shall be cycled between 5 to 500 cps or as specified in the applicable associated specification and return in a 15 minute cycle at an applied double amplitude of 0.036 inch or an applied acceleration of ±10G which ever is the lower G value. The rate of change of frequency shall be logarithmic (see 3.3.10.3).

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in the contract or order (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 75 liter, 300 psig converter covered by this specification is intended to be used in the C-5, C-141, E-3 and E-4 aircraft.

6..2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-25666 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from, OC-ALC/TICLA, 3001 Staff Drive, Tinker AFB, OK 73145-3037.

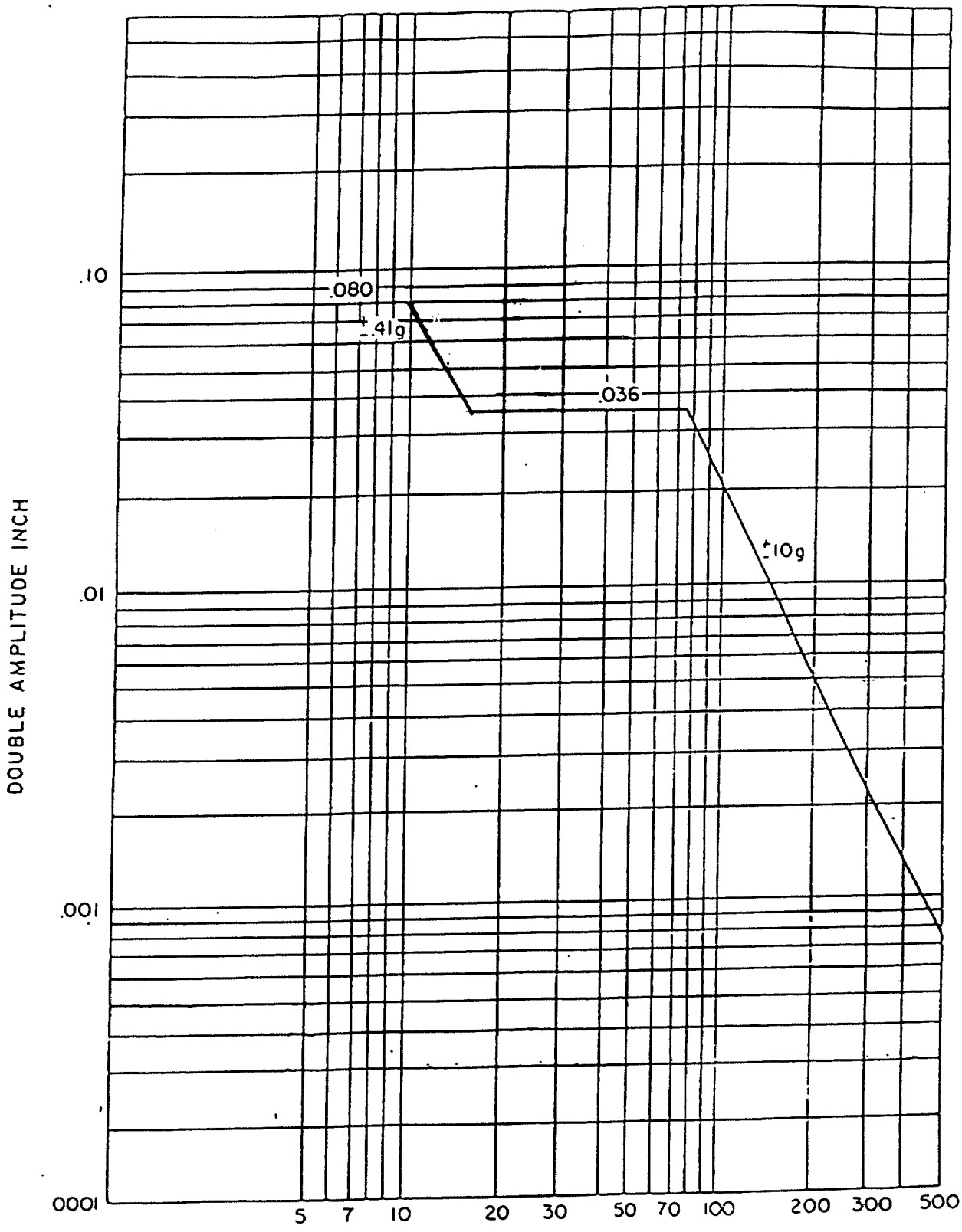
6.4 Subject term (key word) listing.

Breathing oxygen
Capacitance
Container, inner
Container, outer
Delivery rate
Dewar
Element, sensing
Evaporation loss
Gunfire
Life support
LOX
Valve, fill-buildup-vent
Valve, pressure closing
Valve, pressure relief
Vibration

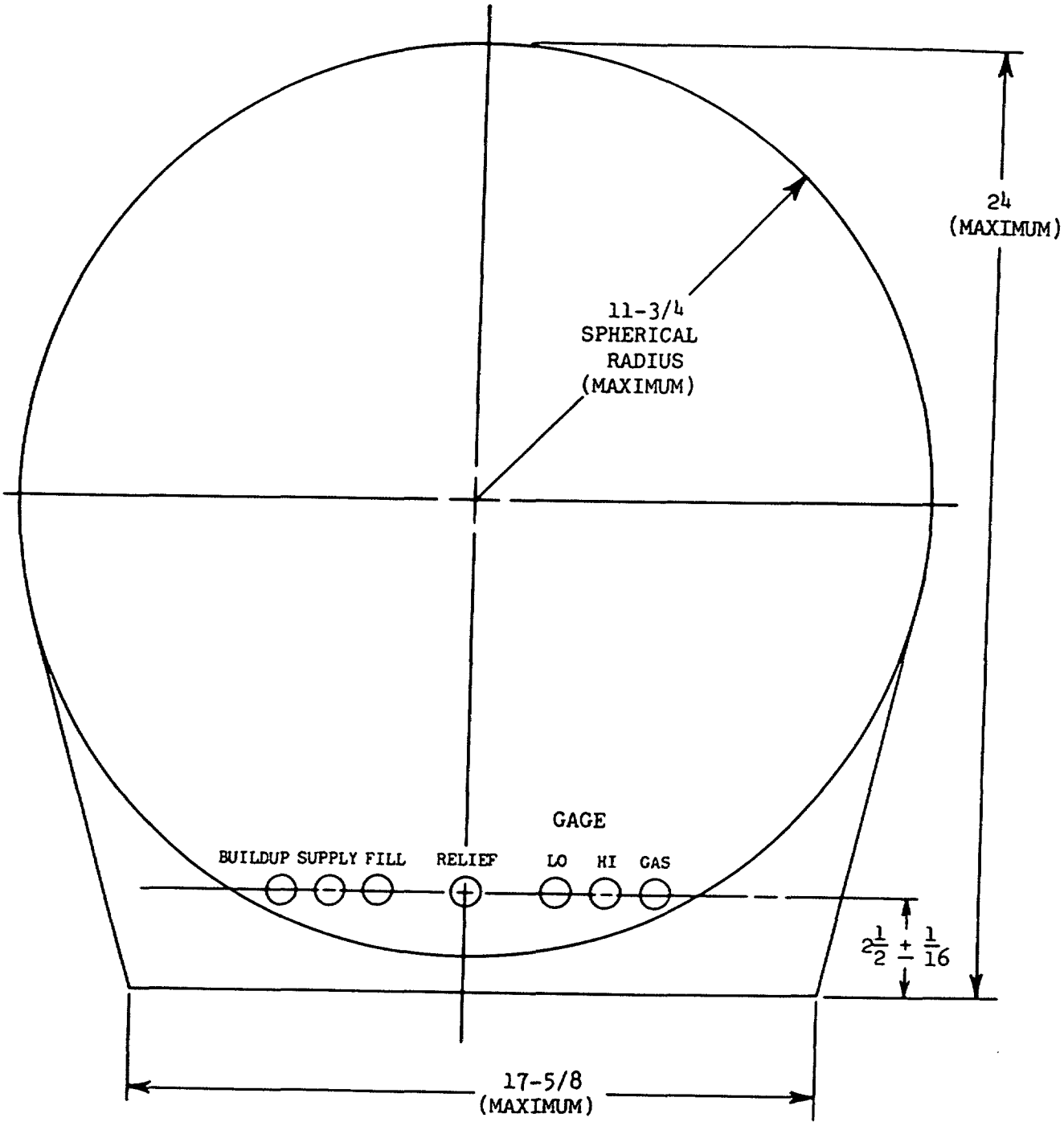
6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE I. Range curve for converter vibration



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DIMENSIONS IN INCHES

TOLERANCES:
FRACTIONS ± 1/64

FIGURE 1. Front view

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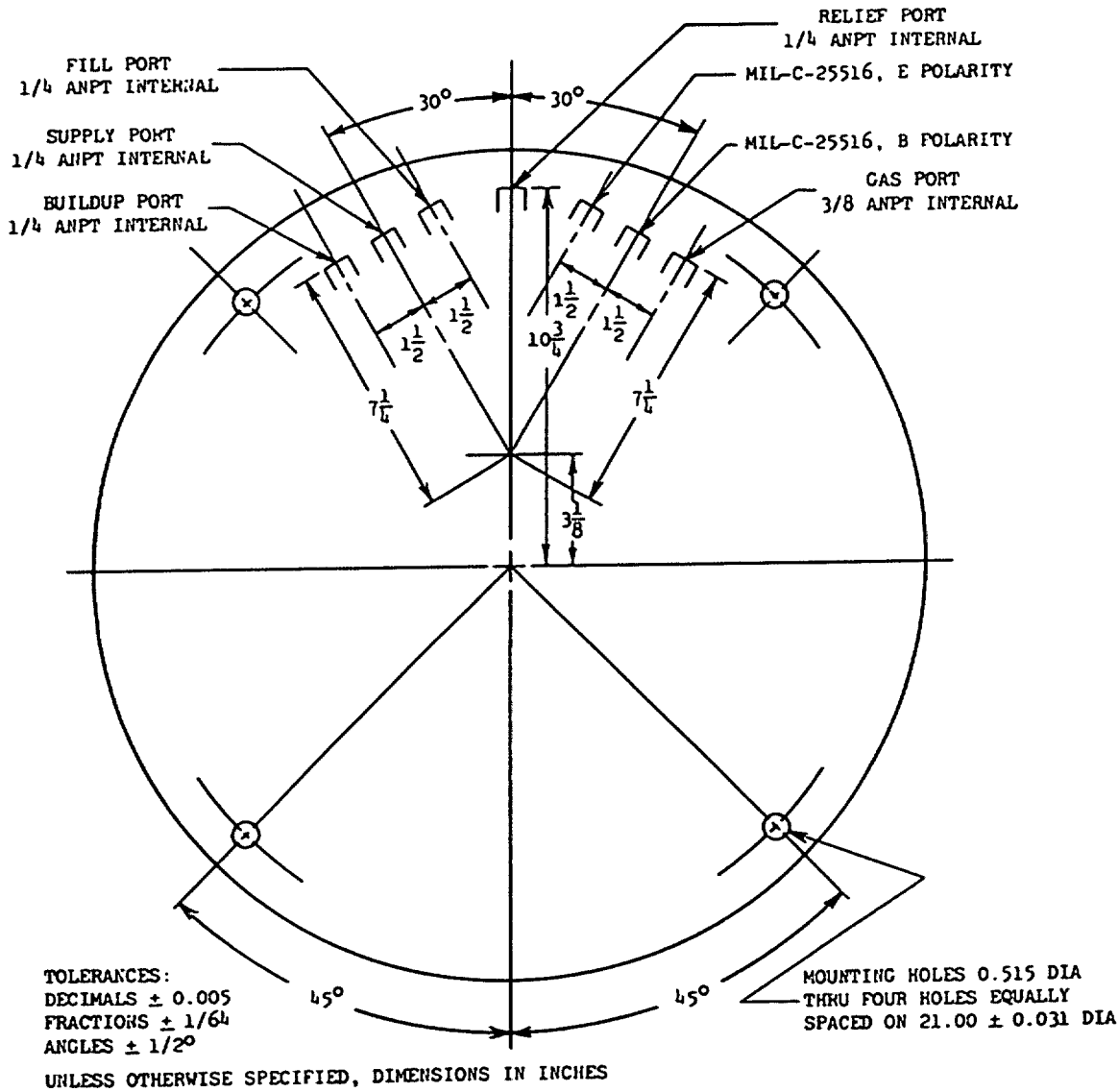


FIGURE 2. Bottom view

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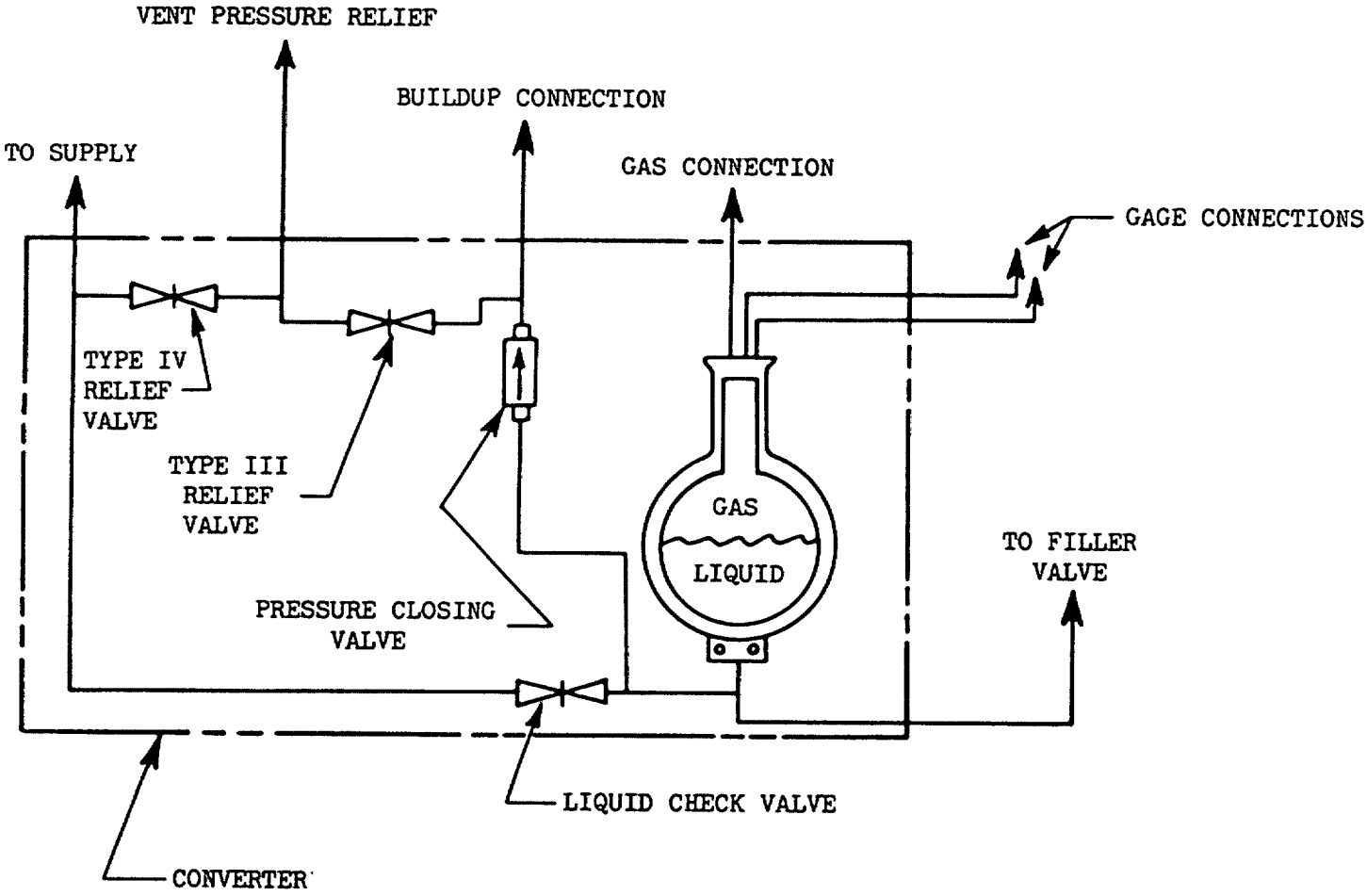
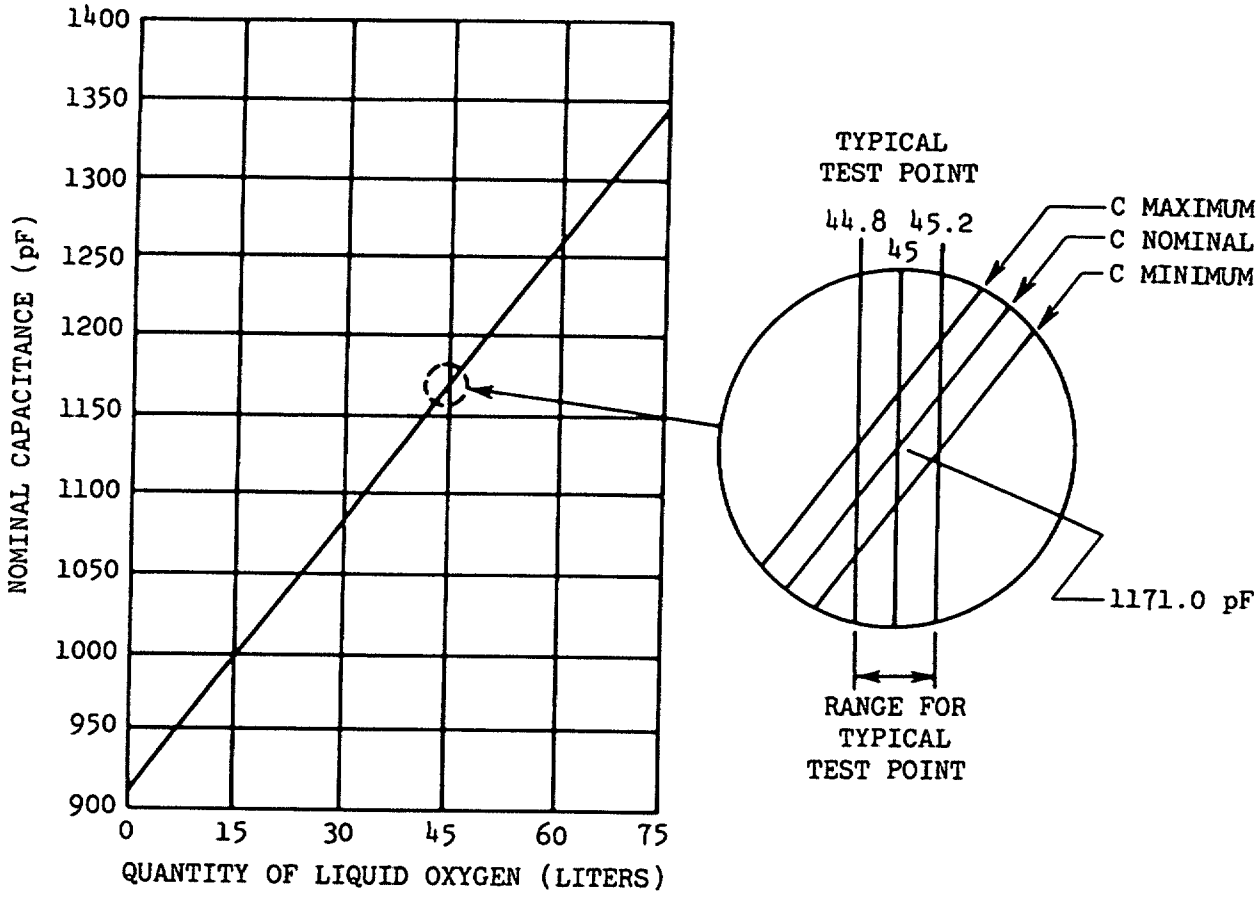


FIGURE 3. Schematic diagram

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Liquid oxygen (liters)	Nominal capacitance (pf)
0	910.0
15	997.0
30	1084.0
45	1171.0
60	1258.0
75	1345.0

FIGURE 4. Relationship between capacitance and volume of oxygen sensed

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-25666/6	950915

3. DOCUMENT TITLE: CONVERTER, LIQUID OXYGEN GCU-20/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

OC-ALC/TICLA ATTN: H. Edwards 3001 Staff Drive, Ste 1AE1/101A Tinker AFB, OK 73145-5990	b. TELEPHONE (Include Area Code) (1) Commercial: (405) 736-5960 (2) AUTOVON: 336-5960	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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